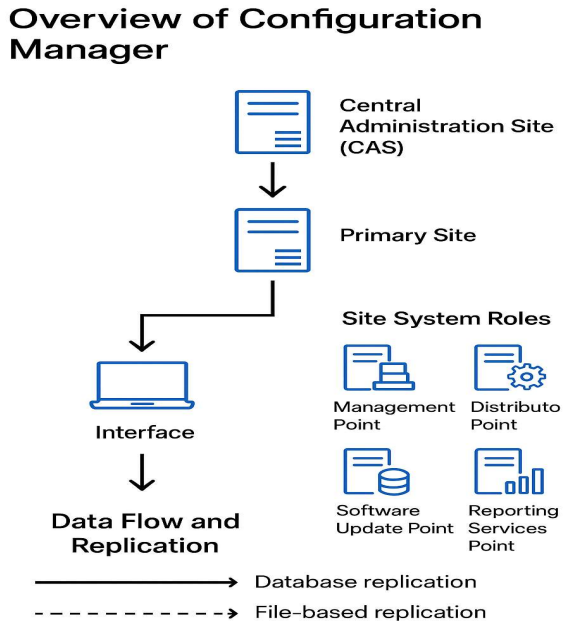
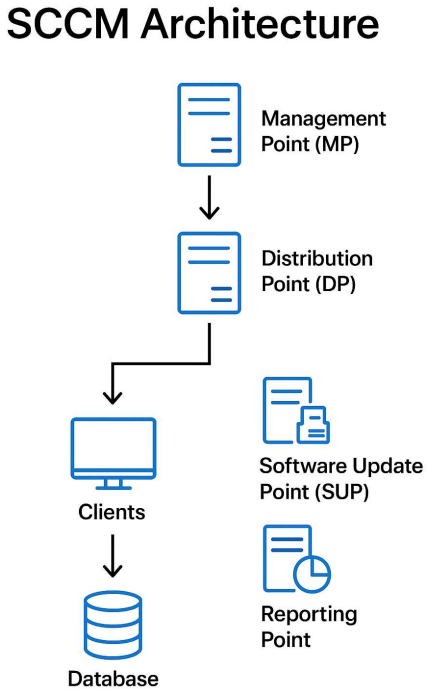
Microsoft SCCM
26 August 2025 11:50

SCCM , SCOM, SC VMM

SCCM integration is with MS SQL and hence we get SSRS [SQL Server Reporting Service] support for reporting.





Here's a clear **overview of Microsoft System Center Configuration Manager (SCCM)**:

◈ What is SCCM?

Microsoft System Center Configuration Manager (SCCM) — now known as **Microsoft Endpoint Configuration Manager (MECM)** — is a **systems management software** that helps IT administrators manage large groups of Windows-based (and also Linux, macOS, iOS, Android) systems.
It is widely used in enterprises to **deploy applications, manage updates, secure devices, and monitor system health** across the network.

◈ Key Functions of SCCM

1. **Software Deployment & Distribution**
   - Automates application installation and updates.
   - Supports MSI, EXE, Windows Store apps, and scripts.
   - Can push software silently to targeted devices.
2. **Operating System Deployment (OSD)**
   - Automates Windows OS installation (bare-metal or upgrade).
   - Supports PXE boot, boot media, and task sequences for customized builds.

3. **Patch Management**
   - Integrates with **Windows Server Update Services (WSUS)**.
   - Ensures devices get latest security patches and feature updates.
   - Allows controlled testing and phased rollout.
4. **Hardware & Software Inventory**
   - Collects details about installed apps, OS versions, hardware configuration.
   - Provides reporting for compliance, licensing, and planning.
5. **Endpoint Protection**
   - Integrates with **Microsoft Defender Antivirus**.
   - Provides malware protection, security baselines, and compliance policies.
6. **Compliance & Settings Management**
   - Enforces configuration baselines (password policy, firewall rules, etc.).
   - Ensures systems meet corporate IT policies.
7. **Remote Control & Monitoring**
   - Allows IT admins to remotely connect to users' devices.
   - Helps in troubleshooting without physical access.
8. **Integration with Microsoft Intune (Co-management)**
   - Hybrid management of on-premises + cloud devices.
   - Useful in modern workplace scenarios with remote/hybrid workers.

◈ SCCM Architecture Components

1. **Site Server** – The central server hosting SCCM roles.
2. **Primary Site** – Manages clients directly.
3. **Secondary Site** – Optional, used for bandwidth optimization in branch offices.
4. **Management Point (MP)** – Main communication point between clients and SCCM.
5. **Distribution Point (DP)** – Stores content (apps, OS images, patches) for client access.
6. **Software Update Point (SUP)** – Integrates WSUS for patching.
7. **SQL Server Database** – Stores all SCCM configuration and inventory data.

◈ Benefits of Using SCCM

- Centralized IT management for thousands of devices.
- Reduces manual effort in software/patch deployment.
- Enhances security compliance.
- Scales from small organizations to very large enterprises.
- Supports both **traditional on-premises management** and **modern cloud co-management** with Intune.

✅ In short: **SCCM is a powerful enterprise management tool** that gives IT admins centralized control over devices, applications, security, and compliance across the organization.

◈ New Features on SCCM (Now MECM)

Microsoft continuously updates SCCM (part of **Microsoft Endpoint Manager**). Some **recent/new features** include:

- **Tenant Attach with Intune** → Manage on-prem SCCM devices directly from the Intune portal.
- **Cloud Management Gateway (CMG)** → Manage internet-based clients without VPN.
- **Improved Co-management** → Integration with Microsoft Intune for hybrid device management.
- **Third-party patching support** → Deploy non-Microsoft updates.
- **Application Groups** → Deploy multiple apps as a single group.
- **CMPivot** → Real-time query tool for client health, installed software, running processes.
- **Improved OS Deployment** → Task sequence improvements, peer cache for faster OSD.
- **Endpoint Analytics Integration** → Device performance and reliability insights.

◈ Benefits of Implementing SCCM in an Organization

1. **Centralized Management** → One console for applications, OS, updates, and compliance.
2. **Improved Security** → Automated patching + Defender integration.
3. **Time & Cost Savings** → Reduces manual IT work.
4. **Scalability** → Supports 100s to 100K+ devices.
5. **Remote Workforce Support** → CMG & Intune integration.
6. **Compliance & Auditing** → Ensures licensing and policy adherence.

◈ Overview of the Configuration Manager Hierarchy and History

Hierarchy

SCCM can be deployed in different hierarchical models:
- **Standalone Primary Site** → Single server/site managing all clients.
- **Hierarchy with CAS (Central Administration Site)** → Used for very large enterprises (>150K clients). Manages multiple primary sites.
- **Secondary Sites** → Deployed in branch offices to reduce WAN traffic; report to a primary site.

History

- **1994**: Systems Management Server (SMS) introduced.
- **2007**: Rebranded as **System Center Configuration Manager (SCCM)**.
- **2012**: Major redesign with role-based administration.
- **2019+**: Integrated into **Microsoft Endpoint Manager**; renamed **Microsoft Endpoint Configuration Manager (MECM)**.

◈ What Is a Central / Primary / Secondary Site?

- **Central Administration Site (CAS):**

- Top-level site in a hierarchy.
- Manages multiple primaries.
- Does not manage clients directly.
- Used for reporting, global policy, and hierarchy-wide configuration.

- **Primary Site:**
  - Manages clients directly.
  - Stores client data in its database.
  - Can have **secondary sites** below it.
  - Required if you want client management.
- **Secondary Site:**
  - Optional; used to manage WAN traffic in branch offices.
  - Relies on the parent **primary site**.
  - Has no separate admin console; configuration done from the primary.

◈ Configuration Manager Site System Roles (and Changes)

Key **site system roles** include:

- **Management Point (MP):** Client communication hub.
- **Distribution Point (DP):** Stores apps, OS images, patches.
- **Software Update Point (SUP):** Integrates with WSUS for patching.
- **Reporting Services Point (RSP):** Generates reports via SQL Server Reporting Services.
- **Endpoint Protection Point (EPP):** Manages Defender AV policies.
- **Fallback Status Point (FSP):** Helps clients report errors.
- **Service Connection Point (SCP):** Connects SCCM to Microsoft cloud services.

👉 **Recent changes:** More roles consolidated or made optional (e.g., reporting improvements, SUP flexibility, CMG cloud role).

◈ How Data Flows and Replicates in a Hierarchy

1. **Clients** → **Management Point (MP)** → Client sends inventory, requests policies.
2. **MP** → **Site Server/Database** → Stores data in SQL DB.
3. **Primary** ↔ **CAS** → Data replicates via SQL-based replication (not file-based anymore).
4. **Primary** ↔ **Secondary** → File-based replication (content, packages).

There are two types of replication:

- **Database replication (SQL Service Broker)** → Used between CAS and primary sites.
- **File-based replication** → Used between primary and secondary sites.

◈ Monitoring Site and Component Status

- **Monitoring Node (SCCM Console):**
  - Monitors **site health, replication, component status, distribution status**.
  - **Component Status node** → Shows server roles health.
  - **Site Status node** → Monitors overall site health.
  - **Replication Monitoring** → Shows database sync status between sites.

◈ Managing Components Using Service Manager

- **Configuration Manager Service Manager** tool allows admins to:
  - View site components and services (MP, DPs, SUP, etc.).
  - Start/stop individual components.
  - Troubleshoot issues (e.g., restarting SMS Executive components).

◈ What Are Configuration Manager Log Files?

- SCCM has **hundreds of log files** for troubleshooting.
- Located in:
  - **Client logs:** C:\Windows\CCM\Logs
  - **Server logs:** C:\Program Files\Microsoft Configuration Manager\Logs
- Common examples:
  - execmgr.log → Application execution
  - smsts.log → OSD task sequence troubleshooting
  - WUAHandler.log → Windows Updates
  - distmgr.log → Distribution point/package distribution
  - mpcontrol.log → MP health
  - sitecomp.log → Site components installation/config

✅ This gives you a **full breakdown of SCCM: features, hierarchy, site types, roles, replication, monitoring, service manager, and logs**.

Note: Prepared by - K.T. Harsha . Images are taken from the web site, used only for training and understand purpose